Use: Autumn Science: Show Your Thinking

Today I can… ☐ Say how friction helps in one case. Say how it hinders in one case. ☐ Say what we gain from a mechanism. Say what it costs. ☐ For a fair test, say what to change. Say what to measure. Say what to keep the same. ☐ Explain an orbit. Say why we see light from the Moon.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

STAFF ANSWERS · Supported

Arrival

1. Friction helps shoes grip.
2. Earth.
3. Keep the block the same while changing the surface.
4. The Moon reflects sunlight.

Exit and independent answers

Check the pupil’s own evidence against the learning criteria. Say how friction helps in one case. Say how it hinders in one case.; Say what we gain from a mechanism. Say what it costs.; For a fair test, say what to change. Say what to measure. Say what to keep the same.; Explain an orbit. Say why we see light from the Moon.

Keep the first work and the changed work. A claim needs clues that fit and a science reason. Keep the model limit clear. These teaching examples do not show that you did a test.

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STAFF ANSWERS · Standard

Arrival

1. Brakes/shoes: helpful resistance; engine rubbing: unwanted resistance and heating.
2. The fixed point the lever turns around.
3. Same block, ramp, release point and measurement method.
4. Gravity curves Earth’s path; reflected sunlight lights the Moon.

Exit and independent answers

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STAFF ANSWERS · Stretch

Arrival

1. For example, brake pad and disc touch; friction opposes their relative sliding.
2. Less effort force requires more effort distance for the same load rise.
3. Two planned changes prevent isolating which caused the difference.
4. Normal phases reflect viewing geometry; Earth’s shadow causes a lunar eclipse.

Exit and independent answers

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STAFF ANSWERS · teaching cues

Hinge

Use the same block, ramp and point to let go. Measure in the same way. Change just the surface. Measure how far it goes in centimetres.

Model steps and limits

Use genuine work if it is available; a missing record stays missing.

Here are data to think about. The same block went 45 cm on smooth board. It went 19 cm on fabric. These data are here as an example. They are not results from your own test.

The evidence can support a claim about these surfaces and this block; one example is not every surface.

A lever may reduce the effort force by increasing the effort distance. It does not create energy.

Gravity pulls planets inward and bends their paths. Earth is third from the Sun.

The Moon reflects sunlight. Changing view of its lit half creates phases; a lunar eclipse involves Earth’s shadow.

Illustrative data support reasoning practice only. They are never recorded as a completed practical.

Activity answers

Brakes: Friction helps the brakes slow movement.

Lever: A mechanism trades distance for force.

Controls: One planned variable changes.

Moonlight: The Moon does not make its own visible light.

Orbit: Gravity bends the moving planet’s path.

Engine: Friction can hinder a machine while helping brakes.

Misconception repair

In science, a claim needs clues that fit. Give a reason. Say how far the claim can go.